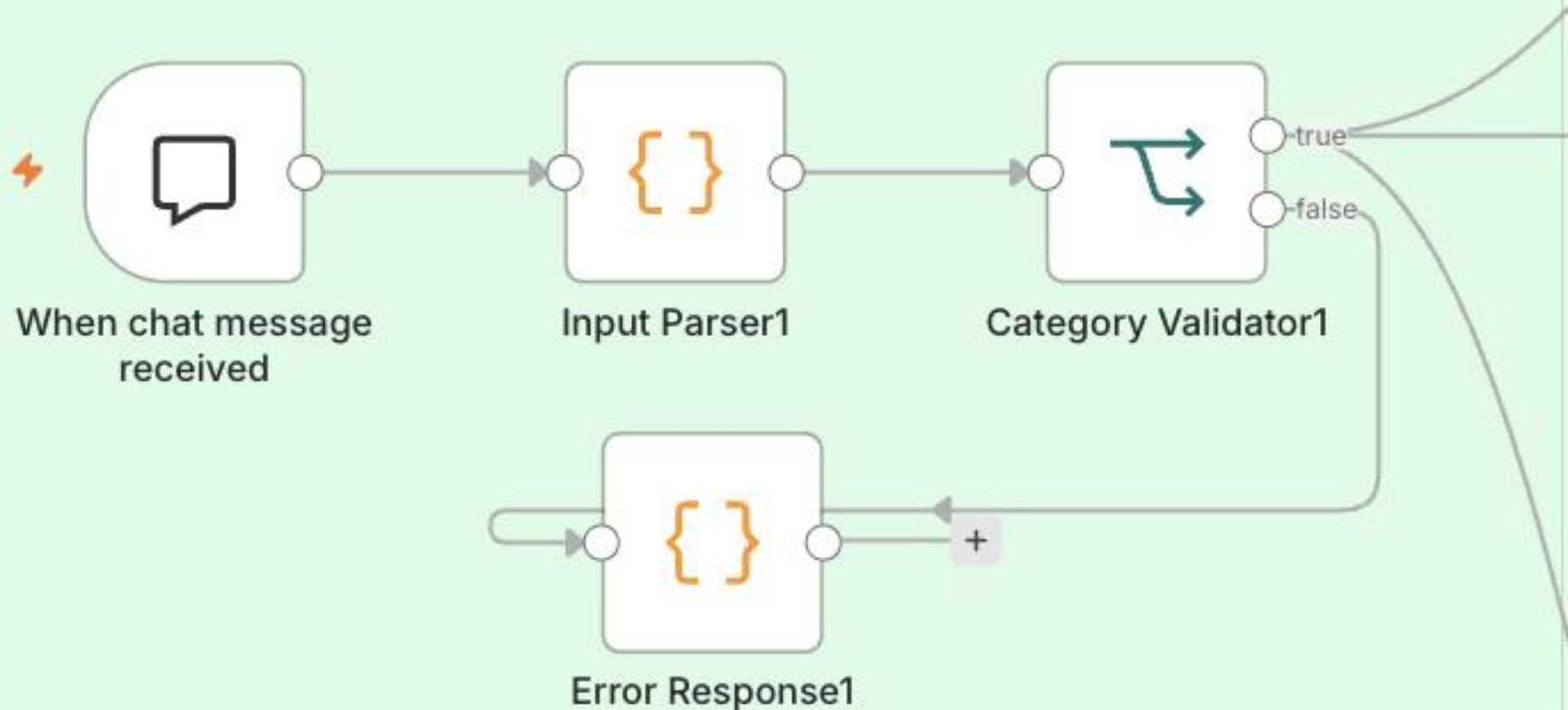


Stage 1: Input & Routing

Parses user input to extract:

- Category Selection (Area Rug, Outdoor Rug, Hallway Runner, Shag Rug)
- **Amazon.com** URLs only (rejects other Amazon domains)
- Product vs. Collection URL Classification

Validates Input before proceeding



Stage 2A: Amazon Scraping

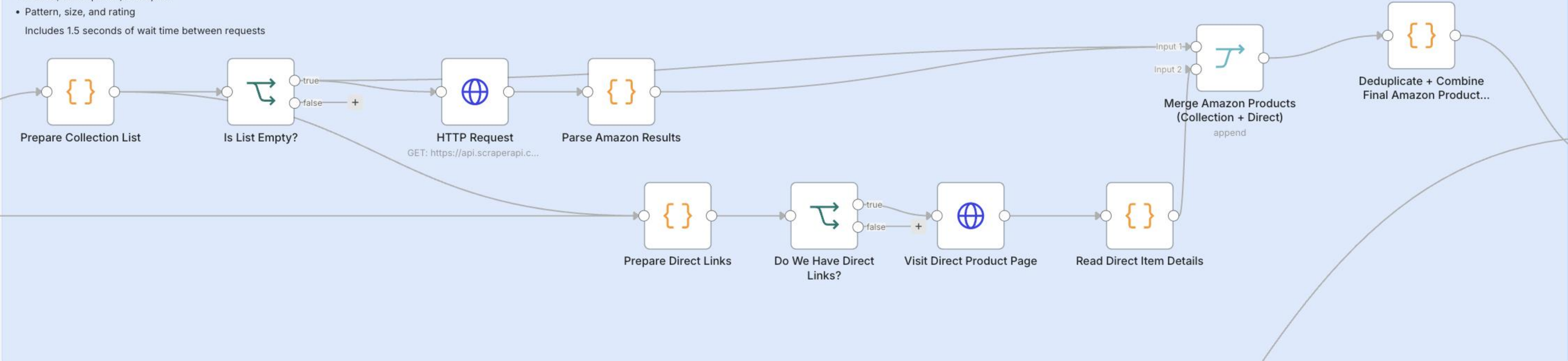
Two Paths:

- 1. Collection URLs ⇒ Extract Product Links ⇒ Scrape each
- 2. Direct product URLs ⇒ Scrape directly

Use Cheerios to parse:

- Product name, price, and brand
- Details, descriptions, and specs
- Pattern, size, and rating

Includes 1.5 seconds of wait time between requests



Stage 2B: Extern API & Image Analysis

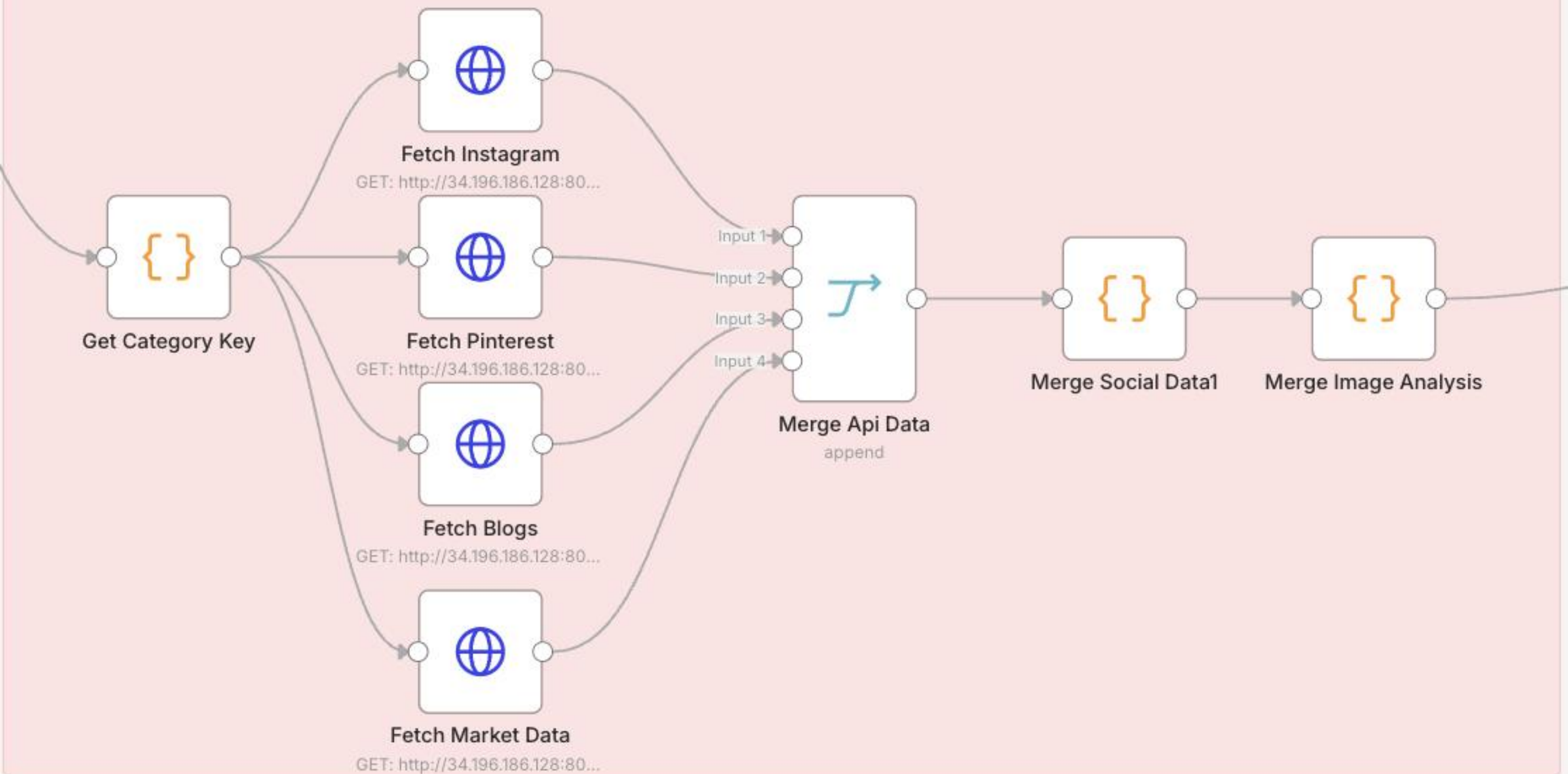
Calls FastAPI server for social data:

- Instagram trends (with images)
- Pinterest boards (with images)
- Blog posts (text only)

New: Gemmini Vision analyzes social images

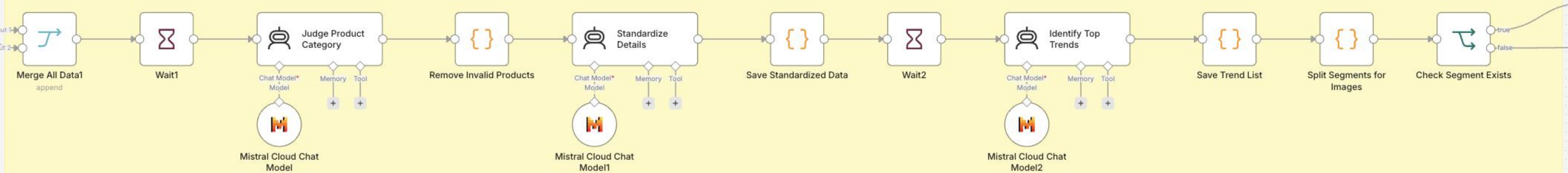
- Extracts colors, patterns, and styles
- Identifies texture and moods
- Provides visual trend summary

Gracefully handles API failures



Stage 3: AI Processing

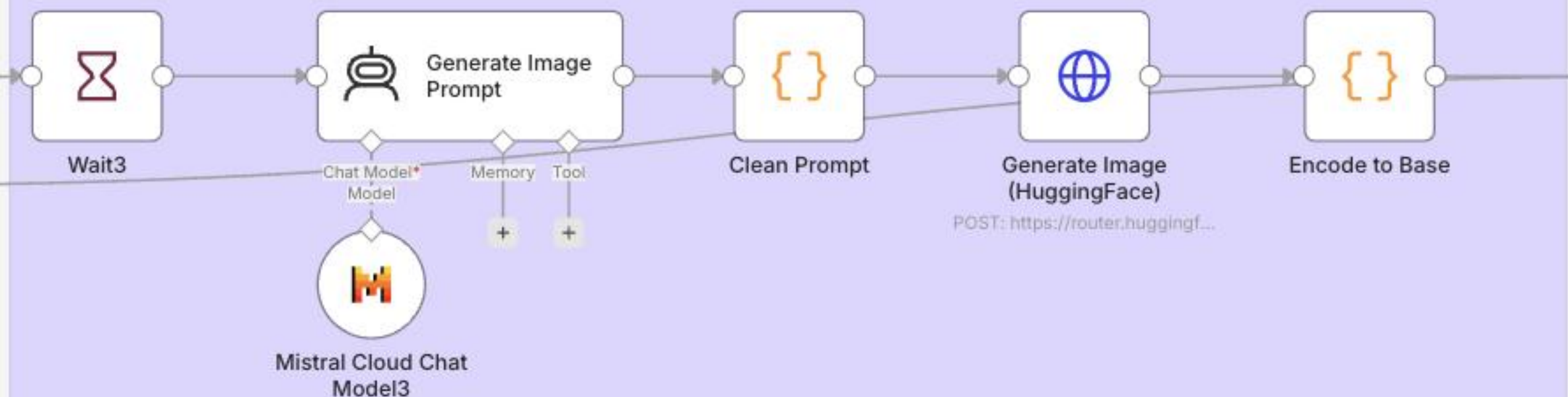
- 1. Category Classifier ⇒ Filters products to match selected categories
- 2. Attribute Extractor ⇒ Normalize size, color, material, pattern, and style
- 3. Micro-Segement Identifier ⇒ Find 2-3 style trends within selected category



Stage 4: Generating Trend Images from AI Micro-Segments

For each micro-segment

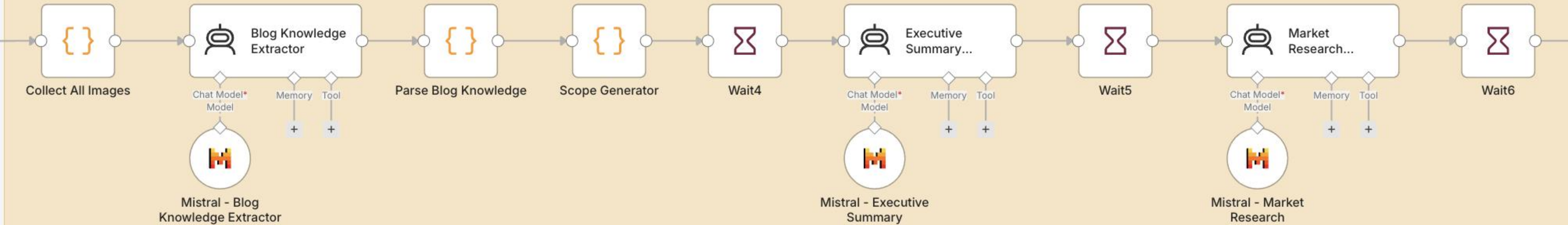
1. Generate optimized image prompt
2. Clean prompt for API
3. Call HuugingFace FLUX.1-schnell
4. Encode to base64 for HTML embedding

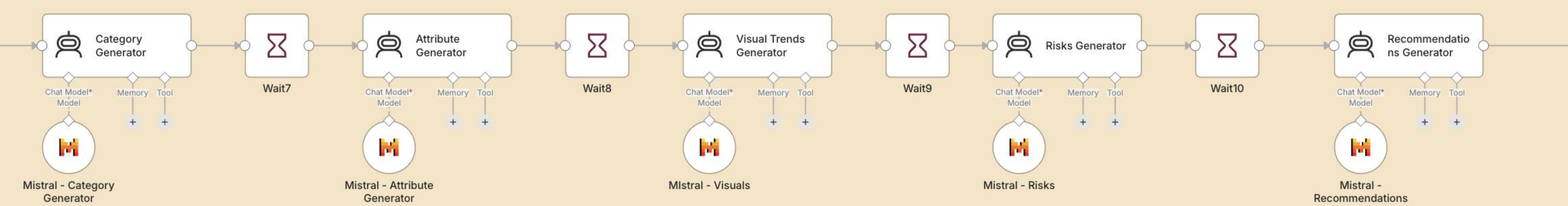


Stage 5-6: Merge & Report

- 1. Convert blog articles into structured expertise (materials, sizing, buyer guidance)
- 2. Use multiple AI writer agents to generate each section of a professional HTML trend report
- 3. Merge all sections, inject images, and produce a single downloadable, styled HTML file

Goal: Turn raw analysis into a polished, shareable market report with visuals and narrative

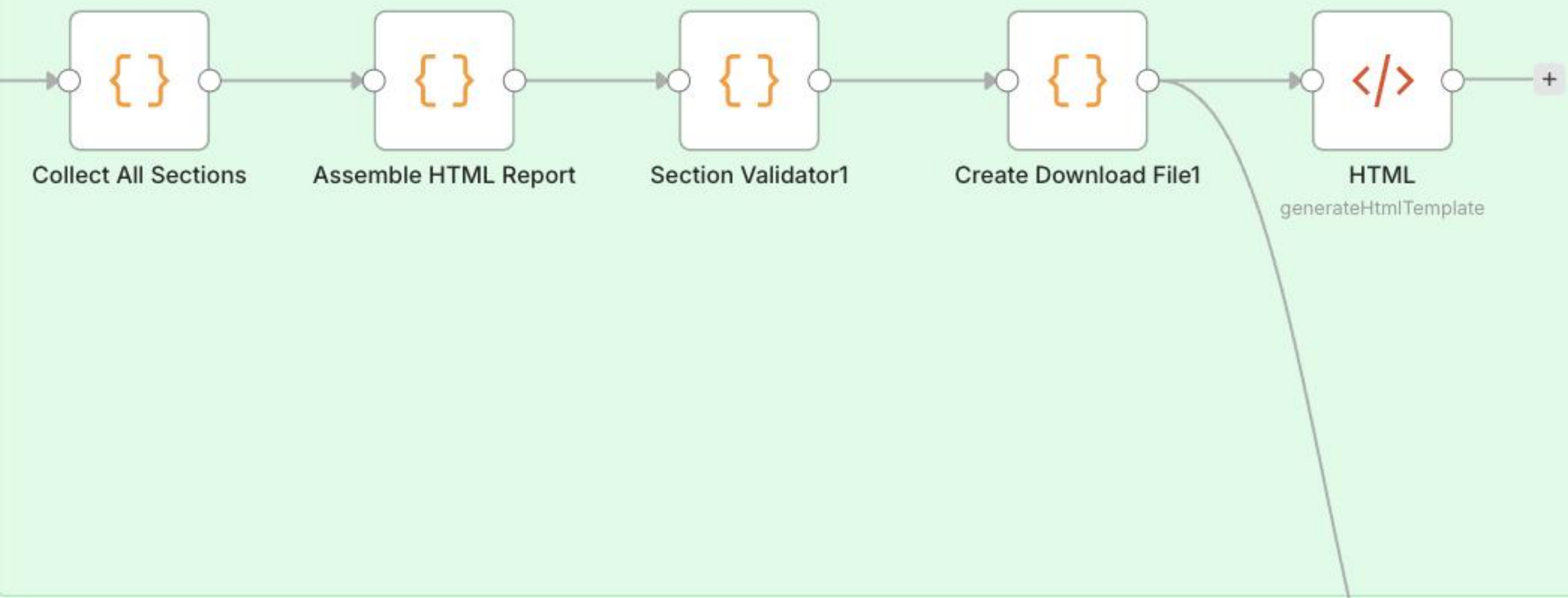




Stage 7: Output

- HTML cleanup & validation
- Section presence check
- Create downloadable files
- Save to Supabase

Download HTML from "Create Dashboard File" node



Stage 8: Upload to Google Drive

